SIMATS ENGINEERING

CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Code of Conduct

I **R. Pranathi(192525076)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Pranathi

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. **Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.**

ANS:

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Runs directly on hardware	Runs on a host operating system
Performance	Very high, low overhead	Lower due to host OS overhead
Security	More secure with smaller attack surface	Less secure because host OS can be attacked
Manageability	Supports centralized VM management	Suitable for small-scale testing
Examples	VMware ESXi, Microsoft Hyper-V, Xen	VMware Workstation, Oracle VirtualBox

JUSTIFYING THE BETTER CHOICE:

TYPE-1 HYPERVISOR IS THE BEST CHOICE Because:

- High performance
- Strong security
- High availability
- Better scalability

2. **A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.**

ANS:

- Each department (Finance, Healthcare, Student Portal) runs inside a separate Virtual Machine.
- Every VM has its own operating system, CPU, memory, storage, and network.
- The hypervisor prevents the one VM from accessing another VM's resources.
- Virtual networks and storage policies separates the workloads.

Two Major Attack Surfaces

1. Hypervisor Attack

- Attackers exploit vulnerabilities in the hypervisor.
- May gain control over multiple VMs.
- Regular patching, secure configuration, least-privilege access.

2. VM Escape Attack(Virtual Machine(VM))

- Malware inside one VM escapes to the host or other VMs.
- Can compromise sensitive data.
- Strong isolation, updated hypervisor, endpoint security.

3. **Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.**

ANS:

Given,

- CPU = 92%
- Memory = 88%
- Storage I/O Latency = 35 ms
- VM Boot Time = 170 seconds

Bottlenecks:

CPU (92%)

- CPU is nearly overloaded.
- VMs compete for processing power.
- Causes slow application performance.

Memory (88%)

- High memory usage may trigger swapping.
- Leads to slower VM execution.

Storage I/O Latency (35 ms)

- Storage is responding slowly.
- Disk operations become delayed.
- VM startup time increases.

VM Boot Time (170 s)

- Boot time is much higher than normal.
- Indicates storage and resource congestion.

AND ALSO:

- ➢ Increase RAM capacity.
- ➢ Upgrade storage
- ➢ Remove unused VMs and optimize resource allocation.

4. **Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.**

ANS:

Feature	Traditional Datacenter	Software Defined Datacenter
Compute	Physical servers	Virtual machines
Storage	Dedicated storage	Software-defined storage
Network	Physical switches	Software-defined networking
Provisioning	Manual	Automated
Scalability	Slow	Fast

Virtualization

- It is connected Multiple VMs to one physical servers.

Storage Virtualization

- It Combines multiple storage devices into one logical pool.
- It Improves utilization and flexibility.

Network Virtualization

- Creates virtual networks independent of physical hardware.
- Enables rapid deployment and isolation.

How Software Defined Datacentre Improves Agility

- Faster deployment of services.
- Automated resource allocation.
- Easy scalability.
- Low cost.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

ANS:

Analysing the Problem:

Different cloud providers use different identity systems, causing:

- Authentication failures
- Authorization mismatches
- Duplicate user accounts
- Inconsistent access policies

Secure Identity and Privacy Design:

Identity Federation

- Use a centralized Identity Provider (IdP).
- Users authenticate once (Single Sign-On).

Federation Protocols

- SAML
- OpenID Connect

Privacy Protection:

- Share only required user information.
- Encrypt the identity data during the transmission.
- Uses Multi-Factor Authentication (MFA).
- Apply the Role-Based Access Control (RBAC).

Benefits:

- Single login across all cloud providers.
- Consistent user identity.
- Improved security.
- Better privacy protection.
- Simplified access management.